DAY — 07 SEAT NUMBER | | | | | | |

2018	11	28	1100	J - 255	(E)

CHEMISTRY (55)

Time : 3 Hrs. (7 Pages) Max. Marks : 70

Notes :

(i) All questions are compulsory.

(ii) Answers to the questions of Section - I and Section - II should be written in the same answer book.

(iii) Draw neat, labelled diagrams and write balanced chemical equations wherever necessary.

(iv) Figures to the right indicate full marks.

(v) Use of logarithmic table is allowed.

(vi) Answer to every new question must be started on a new page.

SECTION - I

Q. 1. Select and write the most appropriate answer from the given alternative for each sub-question : [7]

(i) The process in which the value of $\Delta U = 0$ is –

(a) Adiabatic (b) Isothermal

(c) Isobaric (d) Isochoric

(ii) An ionic crystal lattice has $\frac{r}{r}$ radius ratio of 0·320, its co-ordination number is –

(a) 3 (b) 4

(c) 6 (d) 8

https://byjus.com

(iii) In hydrogen-oxygen fuel cell the carbon rods are immersed in hot aqueous solution of –

(a) KCl (b) KOH

(c) H_2SO_4 (d) NH_4Cl

(iv) The chemical formula of willemite is –

(a) ZnS (b) $ZnCO_3$

(c) ZnO (d) Zn_2SiO_4

(v) The oxidation state of nitrogen in dinitrogen trioxide is –

(a) +1 (b) +2

(c) +3 (d) +4

(vi) Which of the following 0·1M aqueous solutions will exert highest osmotic pressure?

(a) $Al_2(SO_4)_3$ (b) Na_2SO_4

(c) $MgCl_2$ (d) KCl

(vii) The half-life period of zero order reaction A $\rightarrow$ product is given by –

(a) $\frac{[A]_0}{k}$ (b) $\frac{0 \cdot 693}{k}$

(c) $\frac{[A]_0}{2k}$ (d) $\frac{2[A]_0}{k}$

Q. 2. Answer any SIX of the following : **[12]**

(i) Derive the relation between elevation of boiling point and molar mass of solute.

(ii) State third law of thermodynamics. Give 'two' uses.

(iii) Draw a neat and labelled diagram of lead storage battery.

(iv) Ionic solids are hard and brittle. Explain.

https://byjus.com

(v) A certain reaction occurs in the following steps –

(i) $Cl_{(g)} + O_{3(g)} \rightarrow ClO_{(g)} + O_{2(g)}$

(ii) $ClO_{(g)} + O_{(g)} \rightarrow Cl_{(g)} + O_{2(g)}$

(a) What is the molecularity of each of the elementary steps?

(b) Identify the reaction intermediate and write the chemical equation for overall reaction.

(vi) Define : (a) Semipermeable membrane

(b) Reference electrode

(vii) What is the action of chlorine on :

(a) CS_2

(b) E cess NH_3

(viii) Write the chemical equations involved in van Arkel method for refining zirconium metal.

[9]

Q. 3. Answer any THREE of the following :

(i) Write balanced chemical equations for the following :

(a) Phosphorus reacts with magnesium.

(b) Flowers of sulphur boiled with calcium hydroxide.

(c) Action of ozone on hydrogen peroxide.

(ii) The density of iron crystal is 8·54 gram cm^{-3}. If the edge length of unit cell is 2·8 A° and atomic mass is 56 gram mol^{-1}, find the number of atoms in the unit cell.
(Given : Avogadro's number = $6{\cdot}022 \times 10^{23}$,
$1A° = 1 \times 10^{-8}$ cm)

(iii) How many faradays of electricity are required to produce 13 gram of aluminium from aluminium chloride solution?
(Given : Molar mass of Al = 27·0 gram mol^{-1})

P.T.O

https://byjus.com

(iv) Calculate the internal energy at 298K for the formation of one mole of ammonia, if the enthalpy change at constant pressure is $-42{\cdot}0$ kJ mol^{-1}.
(Given : $R = 8{\cdot}314$ J K^{-1} mol^{-1})

Q. 4. (i) Define : (a) Enthalpy of atomization [7]
(b) Enthalpy of vaporization

(ii) Draw the structure of IF_7. Write its geometry and the type of hybridization.

(iii) (a) State Henry's law.

(b) $22{\cdot}22$ gram of urea was dissolved in 300 grams of water. Calculate the number of moles of urea and molality of the urea solution.
(Given : Molar mass of urea = 60 gram mol^{-1})

OR

(i) What is the action of carbon on the following metal oxides :

(a) Fe_2O_3 in blast furnace

(b) ZnO in vertical retort furnace

(ii) Write the molecular and structural formulae of :

(a) Thiosulphuric acid

(b) Dithionous acid

(iii) The reaction $A + B \rightarrow$ products is first order in each of the reactants.

(a) How does the rate of reaction change if the concentration of A is increased by factor 3?

(b) What is the change in the rate of reaction if the concentration of A is halved and concentration of B is doubled?

https://byjus.com

BYJU'S
The Learning App

SECTION - II

Q. 5. Select and write the most appropriate answer from the given alternatives for each sub-question : [7]

(i) A polymer used in paints is –

(a) omex (b) Thiokol

(c) Saran (d) Glyptal

(ii) The number of primary and secondary hydroxyl groups in ribo e are –

(a) 1, ? (b) 2, 3

(c) 3, 1 (d) 3, 2

(iii) The ligand diethylene triamine is –

(a monodentate (b) bidentate

(c) tridentate (d) tetradentate

(iv) Propene on oxidation with diborane in presence of alkaline hydrogen peroxide gives –

(a propan–1– ol (b) propan–2– ol

(c) allyl alcohol (d) propan–1, 2–diol

(v) Baeyer's reagent is –

(a) acidified potassium dichromate

(b) alkaline potassium dichromate .

(c) alkaline potassium permanganate

(d) acidified potassium permanganate

(vi) Identify 'A' in the following reaction –

$$A + 2Na \xrightarrow[\text{ether}]{\text{Dry}} 2,2,5,5-\text{Tetramethylhexane} + 2NaBr.$$

(a) 2-Bromo-2-methylbutane

(b) 1-Bromo-2, 2-dimethylpropane

(c) 1-Bromo-3-methylbutane

(d) 1-Bromo-2-methylpropane

https://byjus.com

(vii) An antifertility drug is –

(a) Novestrol (b) Histamine

(c) Veranal (d) Equanil

Q. 6. Answer any SIX of the following : **[12]**

(i) Write balanced chemical equations for the conversion of CrO_4^{2-} to $Cr_2O_7^{2-}$ in acidic medium and $Cr_2O_7^{2-}$ to CrO_4^{2-} in basic medium.

(ii) Explain the geometry of $\left[Co(NH_3)_6\right]^{3+}$ on the basis of hybridisation.
(Z of Co = 27)

(iii) Why ethanol has higher boiling point than ethane?

(iv) Write only reactions for the preparation of benzophenone from benzonitrile.

(v) What is the action of p-tolunesulphonylchloride on ethylamine and diethylamine?

(vi) What are amino acids? Write the correct reaction for formation of peptide bond between amino acids.

(vii) Define :

(1) Antiseptics

(2) Antioxidants

(viii) Explain only reaction mechanism for alkaline hydrolysis of tert-butylbromide.

Q. 7. Answer any THREE of the following : **[9]**

(i) Complete and rewrite the balanced chemical equations –

(a) Chlorobenzene $\xrightarrow[473K,\ \text{pressure}]{NaCN + CuCN}$?

https://byjus.com

(b) Isobutyraldehyde $\xrightarrow{50\%\ KOH}$?

(c) Butanone + 2, 4 dinitro-phenyl hydrazine $\xrightarrow{H^+}$?

(ii) Prepare carbolic acid from benzene sulphonic acid.
Write a chemical equation for the action of neutral ferric chloride on phenol.

(iii) Explain the preparation and uses of nylon–2–nylon–6.

(iv) How glucose is prepared from cane sugar?
Write the formula of the complex copper (II) hexacyano ferrate (II).

Q. 8. What is lanthanide contraction? [7]

Explain the cause of lanthanide contraction.

Draw the structures of chloroxylenol and adenine.

How are ethylamine and ethyl methyl amine distinguished by using nitrous acid?

OR

What is the action of the following reagents on ethanoic acid?

(a) $LiAlH_4/H_3O^+$

(b) PCl_3, heat

(c) P_2O_5, heat

Identify 'A' and 'B' in the following reaction and rewrite the complete reaction :

$$CH_3 - CH_2 - Br + AgCN \xrightarrow{\Delta} A \xrightarrow[C_2H_5OH]{Na} B$$

Explain Hoffmann bromamide degradation reaction.

◆◆◆

https://byjus.com